

Eigenvalues and eigenvectors

A non-zero vector v is an eigenvector of a matrix A when

$$A v = \lambda v$$

for some scalar λ , called the eigenvalue belonging to v .

Geometrically: A stretches v without turning it. Every other vector is turned; the eigenvectors are the directions that survive.

Worked example

Take $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Its eigenvalues are $\lambda = 3$ and $\lambda = 1$, with eigenvectors $(1, 1)$ and $(1, -1)$ respectively.

Check: $A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Exercises

1. Find the eigenvalues of $\begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}$.
2. Show that a matrix and its transpose share eigenvalues.
3. When does a 2 by 2 matrix have no real eigenvalues?